

Rayleigh Injection Recursions for Dynamic Packet Energy

A Gap-Free Rank-Staircase Bootstrap

Liang Wang

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Abstract

RH-86 introduced a trace-normalized late-memory Gramian whose clock-rank packet captures terminal postblock energy even when principal-angle perturbation bounds are unusable. This paper derives the scalar recursion needed to turn that construction into an all-level proof target.

Let $\hat{G}_j = X_j^* X_j / \|X_j\|_2^2$, $G_j = \hat{G}_j + \eta G_{j-1}$, and let $E_{j,r}$ be the optimal rank- r tail energy of G_j . For a nondecreasing rank staircase r_j and the previous optimal packet P_{j-1} , the rank-staircase injection recursion is

$$E_{j,r_j} \leq \underbrace{\text{tr}((I - P_{j-1})\hat{G}_j)}_{\iota_j} + \eta E_{j-1,r_{j-1}}.$$

Iteration gives a scalar convolution of the one-step Rayleigh injections ι_j . The current snapshot tail is at most $\sqrt{E_{j,r_j}}$, so a uniform late-time injection estimate replaces both singular-vector matching and spectral-gap control.

For $\eta = 1/512$ and the five archived directional models, 192-bit Arb evaluation certifies the last injection relative norm below 4.95×10^{-4} and captured snapshot energy above 0.99999975. The last injection energy is at most 17.6% of the preceding injection, and every computed recursion inequality is green. A packet available one update earlier predicts the terminal state with relative residual below 1.19×10^{-3} ; at the four finer scales the bound is below 4.1×10^{-6} .

This is a rigorous reduction plus finite-scale evidence, not an all-level injection law. Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: Rayleigh injection; dynamic packet; Ky Fan tail; rank staircase; scalar recursion; interval arithmetic.

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1 The scalar quantity behind packet drift

RH-85 proved that a prefix packet can be propagated to the terminal state [2]. RH-86 replaced a single packet snapshot by the online normalized memory [1]

$$G_j = \hat{G}_j + \eta G_{j-1}, \quad \hat{G}_j = \frac{X_j^* X_j}{\|X_j\|_2^2}.$$

Its leading eigenspace minimizes the weighted relative residual, but directly tracking that eigenspace is ill-conditioned. The missing invariant is the energy injected by the new snapshot outside the old packet.

For a positive trace-class operator G , write

$$E_r(G) = \text{tr } G - \sum_{k=1}^r \lambda_k(G).$$

Let $P_{j,r}$ denote any leading rank- r projection of G_j .

2 Rank-staircase injection recursion

Theorem 2.1 (Rank-staircase injection recursion). *Let r_j be nondecreasing and put $P_{j-1} = P_{j-1, r_{j-1}}$. Then*

$$\boxed{E_{r_j}(G_j) \leq \iota_j + \eta E_{r_{j-1}}(G_{j-1}), \quad \iota_j = \text{tr}((I - P_{j-1})\widehat{G}_j).} \quad (1)$$

Moreover,

$$\frac{\|X_j(I - P_{j, r_j})\|_2^2}{\|X_j\|_2^2} \leq E_{r_j}(G_j). \quad (2)$$

Proof. Because $r_{j-1} \leq r_j$, the old projection is an admissible rank-at-most r_j candidate in Ky Fan's variational principle. Therefore

$$\begin{aligned} E_{r_j}(G_j) &\leq \text{tr}((I - P_{j-1})G_j) \\ &= \text{tr}((I - P_{j-1})\widehat{G}_j) + \eta \text{tr}((I - P_{j-1})G_{j-1}) \\ &= \iota_j + \eta E_{r_{j-1}}(G_{j-1}). \end{aligned}$$

The current normalized snapshot is a nonnegative weight-one summand in the optimal memory residual, giving (2). $\square$

The theorem is gap free. It also allows a rank jump: retaining the old rank- r_{j-1} packet is already admissible at rank r_j , and any extra directions can only improve the bound.

Corollary 2.2 (Scalar convolution corollary). *For $b < j$,*

$$E_{r_j}(G_j) \leq \eta^{j-b} E_{r_b}(G_b) + \sum_{t=b+1}^j \eta^{j-t} \iota_t. \quad (3)$$

If $\iota_t \leq \bar{\iota}$ after b , then the second term is at most $\bar{\iota}/(1 - \eta)$. If the injections decay geometrically, the right side is the convolution of two geometric sequences.

Proof. Iterate (1). The two stated bounds follow by summing the resulting geometric series. $\square$

This formula makes the analytic wall precise: prove a late-time scalar bound for ι_t at the half-logarithmic rank. No individual singular value, principal angle, or coordinate packet must be identified.

3 Five-scale injection audit

The audit uses $r = \lceil H_\sigma \rceil + 2$, $\eta = 1/512$, and packet time $j = \lceil 2M/3 \rceil$. The full memory recursion is computed in binary64. The final injection uses the packet at $j - 1$ and the exact lifted snapshot at j ; its residual is evaluated in 192-bit Arb arithmetic. The same lagged packet is propagated to M and evaluated directly.

The coarse right channel is predictive but not yet in the 10^{-5} regime one update early. The four finer levels are substantially stronger. At all scales the final injection is decreasing relative to the immediately preceding injection, and the scalar recurrence retains positive slack.

σ	worst $\sqrt{\iota_j}$	max ι_j/ι_{j-1}	lagged terminal	max utilization
0.16	4.95×10^{-4}	0.0984	1.19×10^{-3}	0.0071
0.08	2.54×10^{-6}	0.0719	4.02×10^{-6}	0.0344
0.04	2.68×10^{-7}	0.0254	1.43×10^{-6}	0.1281
0.02	5.32×10^{-7}	0.1362	6.55×10^{-7}	0.5276
0.01	2.68×10^{-7}	0.1757	1.75×10^{-7}	0.3284

Table 1: Worst directional result at each scale. Utilization is $E_j/(\iota_j + \eta E_{j-1})$ for the final update.

4 Boundary and next theorem

RH-87 converts the dynamic packet problem into a one-dimensional sequence. The next theorem should estimate ι_j from one-step normalized Gram data, ideally by decomposing it into transported old-packet leakage and a finite-rank source-injection term. A useful all-level statement would give a burn-in $b_\sigma = O(\log^2(1/\sigma))$ and a geometric or polylogarithmic bound on ι_j thereafter.

The exact recursion does not prove that bound. The finite audit does not close the all-level injection law, Stage A1 or Stage A4, a relative fixed-disk determinant, a self-adjoint Hilbert–Polya operator, a zeta-zero identity, or the Riemann Hypothesis.

References

- [1] Liang Wang. Trace-normalized late-memory packets. 2026. RH-86 manuscript.
- [2] Liang Wang. Midblock snapshot packets and prefix-only captured-energy certificates. 2026. RH-85 manuscript.

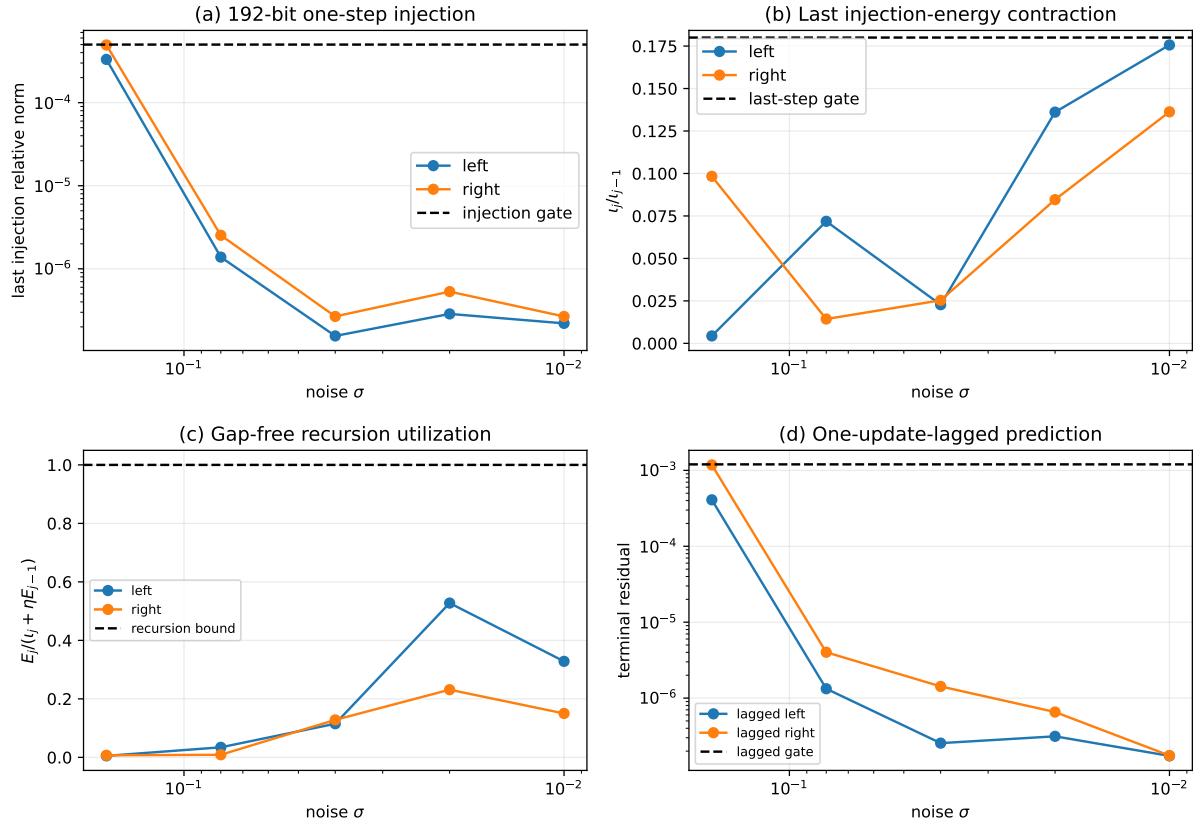


Figure 1: Validated final injection, last-step contraction, recursion utilization, and one-update-lagged terminal prediction.